

AI-Powered Custom Resume Builder

1. Objective

Build an AI-powered platform that automatically creates a **job-specific, ATS-optimized resume** by analyzing a candidate's verified career profile against a target Job Description (JD).

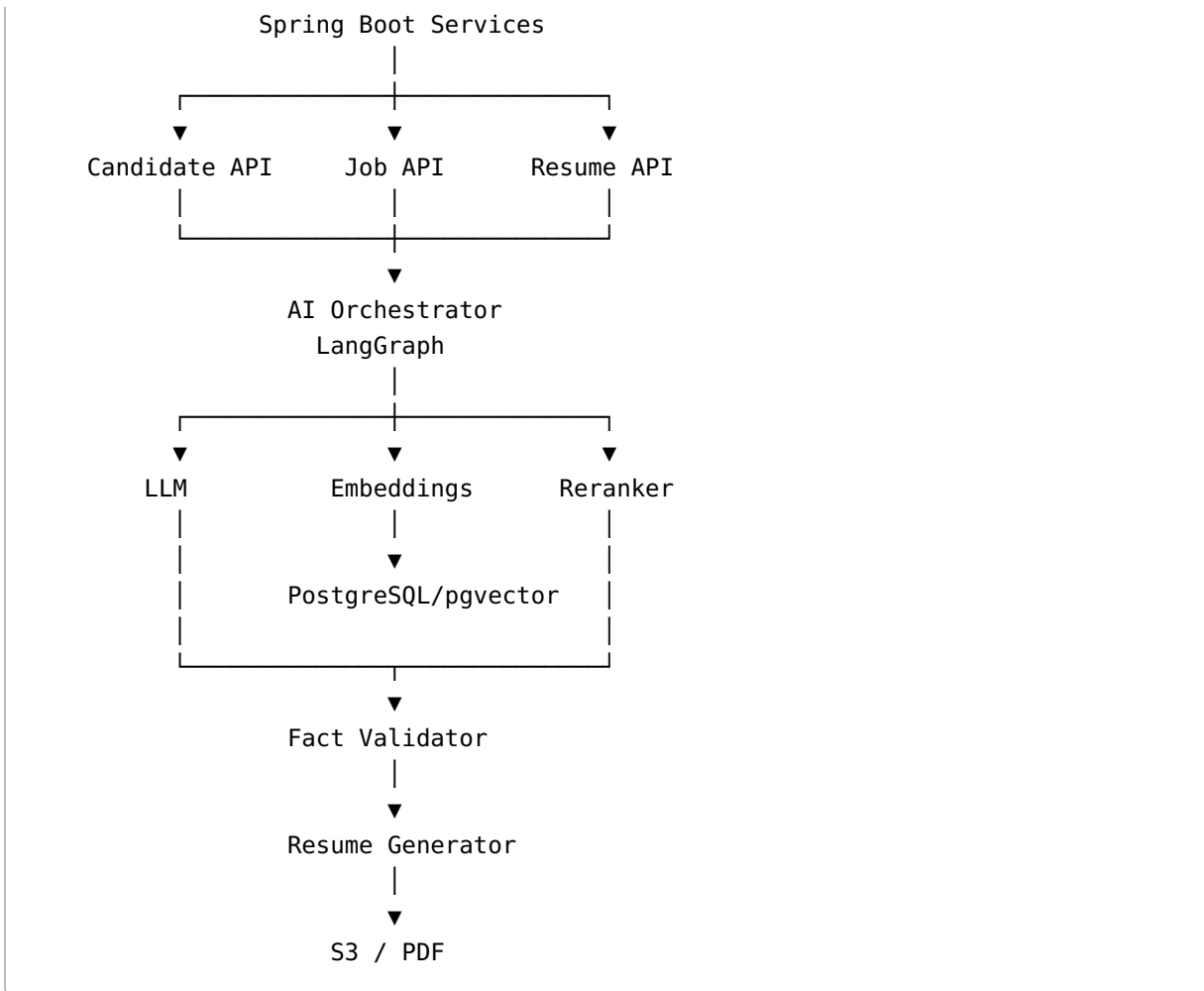
Core principle: AI can optimize wording and highlight relevant experience, but must not invent skills, technologies, employers, certifications, accomplishments, or metrics.

2. User Flow

```
graph TD; A[Master Resume / LinkedIn] --> B[Candidate Career Profile]; B --> C[Paste / Upload Job Description]; C --> D[AI Job Analysis]; D --> E[Candidate ↔ Job Matching]; E --> F[Relevant Experience Retrieval]; F --> G[Tailored Resume Generation]; G --> H[Fact & Hallucination Validation]; H --> I[ATS Optimization]; I --> J[PDF / DOCX Resume];
```

3. Proposed Architecture

```
graph TD; A[Web Application  
React / Next.js] --> B[API Gateway]; B --> C[ ];
```



4. AI Capabilities

Job Description Intelligence

- Extract required/preferred skills
- Identify responsibilities
- Determine seniority and domain
- Extract important ATS keywords

Candidate Intelligence

- Parse resume and LinkedIn profile
- Build structured career profile
- Extract skills, projects and accomplishments
- Maintain evidence/source for every candidate claim

Job Matching

- Keyword matching
- Semantic matching using embeddings
- Experience/relevance matching
- Seniority and domain matching
- Overall Job Match Score

Resume Generation

- Customized professional summary
- Relevant experience selection
- JD-aligned accomplishments
- Skills prioritization
- ATS keyword optimization

Validation

- Verify every generated claim against candidate evidence
- Detect unsupported technologies and metrics
- Prevent fabricated experience
- Flag missing JD requirements rather than falsely adding them

5. Technology Stack

Layer	Proposed Technology
Frontend	React / Next.js / TypeScript
Backend	Java 21 / Spring Boot
AI Service	Python / FastAPI
AI Orchestration	LangGraph
LLM	OpenAI / Claude / Gemini
Enterprise AI Gateway	AWS Bedrock
Embeddings	OpenAI / Gemini
Vector Database	PostgreSQL + pgvector
Database	Amazon Aurora PostgreSQL
Storage	Amazon S3
Messaging	Amazon SQS
Workflow	AWS Step Functions

Layer	Proposed Technology
Authentication	Amazon Cognito
Infrastructure	Terraform
CI/CD	GitHub Actions
Observability	CloudWatch / OpenTelemetry

6. LinkedIn Integration

Use LinkedIn's **OAuth/OpenID Connect** capabilities for user authentication and permitted profile information.

```

User
↓
LinkedIn Authorization
↓
User Consent
↓
LinkedIn Profile Data
↓
LinkedIn Adapter
↓
Candidate Career Profile

```

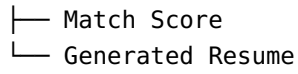
LinkedIn should be an **optional profile enrichment source**, not the application's only source of candidate information. Job descriptions can be supplied through paste, upload, or supported job integrations.

7. Key Data Model

```

Candidate
├─ Employment
│   ├── Projects
│   ├── Achievements
│   └─ Technologies
├─ Skills
├─ Certifications
├─ Education
├─ Resume Versions
└─ Job Applications
    ├── Job Description
    └─ JD Requirements

```



Each experience/achievement should maintain **source, verification status and confidence** to support evidence-based generation.

8. MVP Scope

Phase 1

- Resume upload and parsing
- Candidate career profile
- JD upload/paste
- AI JD analysis
- Candidate-JD matching
- RAG-based experience retrieval
- Customized resume generation
- Fact validation
- ATS optimization
- PDF/DOCX export

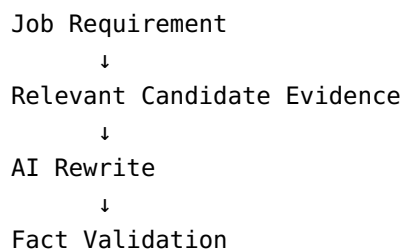
Phase 2

- LinkedIn integration
- Multiple resume templates
- Cover letter generation
- Resume version management
- Application tracking
- Job recommendations
- Interview preparation

9. Key Differentiator

Evidence-backed resume generation.

Every generated resume statement should be traceable to a verified candidate experience.



↓
ATS-Optimized Resume

This creates a **trusted AI career platform**, rather than simply a resume-generation tool.